

PV102 Day 7

Installation Planning Mechanical Layout and JHA

Plan the install before lifting parts.

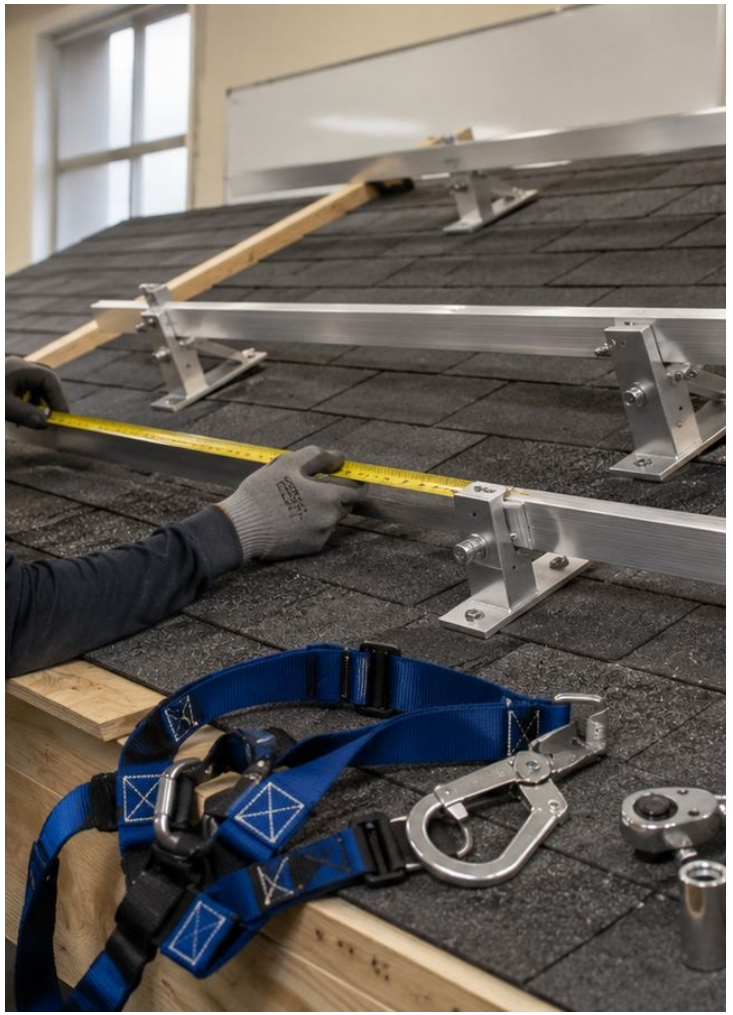
CTS Day 23 | Wednesday, August 12, 2026 | 10:00 AM-3:00 PM

Today’s job

Professional solar work starts before parts move.

• Students plan an installation before touching hardware: layout, material staging, wire routes, access, roof or ground hazards, tool control, quality checks, and job hazard analysis.

- Carry forward the Day 1 habit: match architecture to real need.
- Today adds one design layer and one clean handoff.



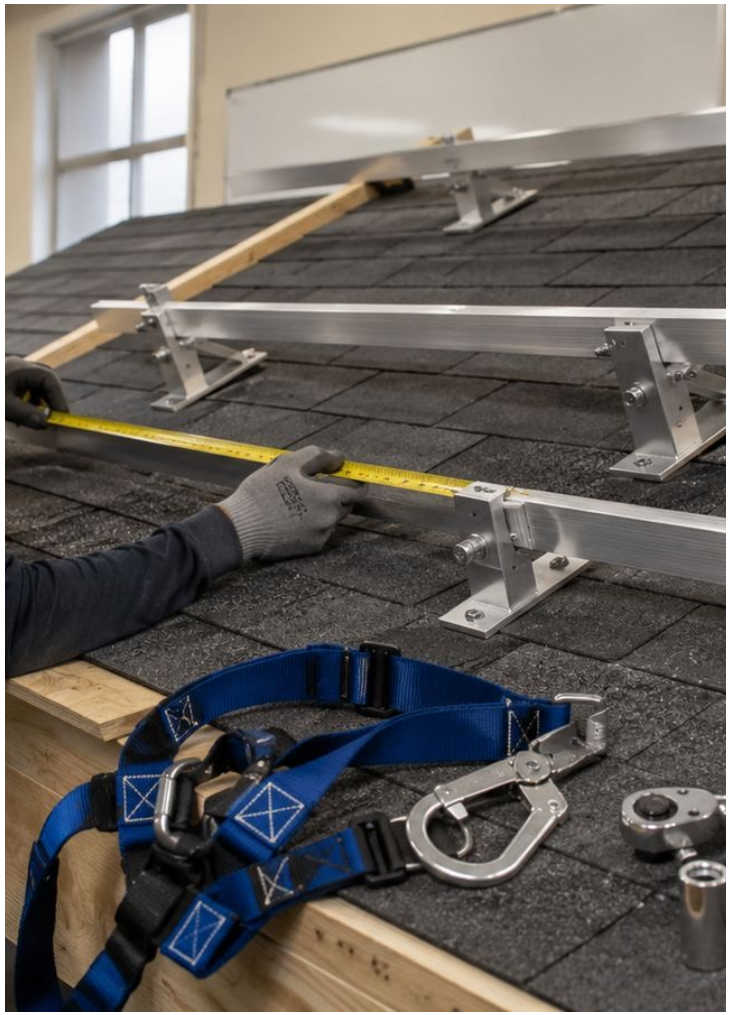
Do not guess when an unknown can be recorded.

Plan the install before lifting parts.

Learning targets

By 3:00 PM students should be able to:

- Convert a design into a sequenced work plan.
- Identify mechanical, electrical, fall, weather, access, and tool hazards.
- Build a material/tool list that matches the scope.
- Use checklists to prevent skipped steps and hidden coupling.



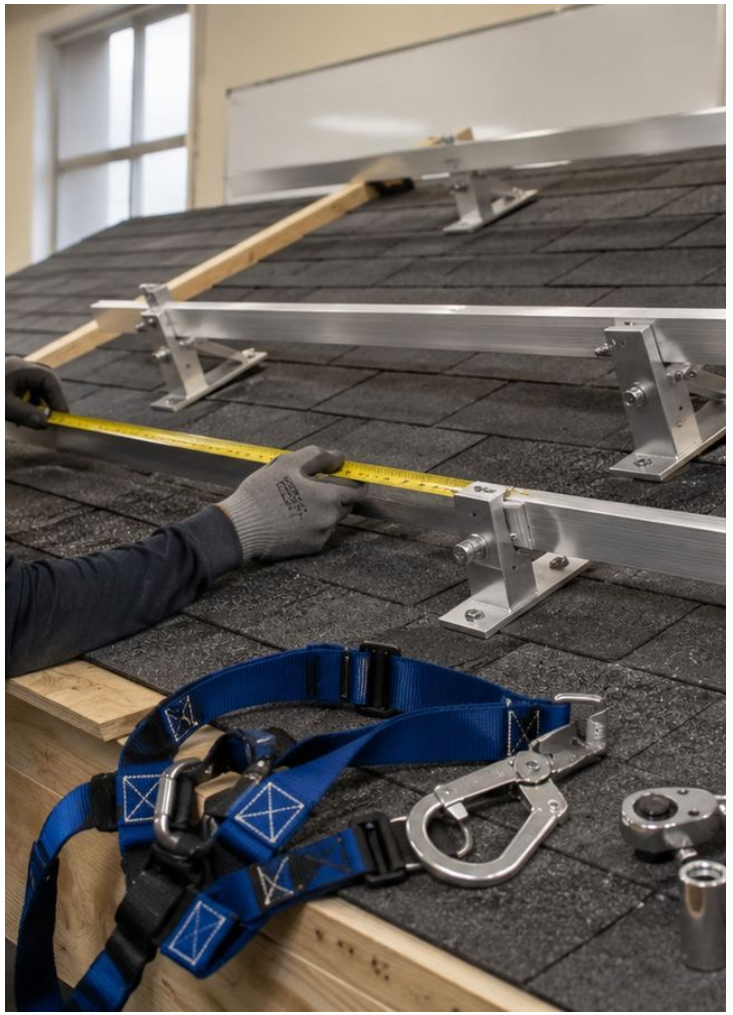
Evidence beats confidence. A tragic lesson for civilization, apparently.

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Morning readiness check

Quiz first, then review as evidence.

- Work independently for 12-15 minutes.
- Use wrong answers to locate teaching gaps.
- Ask: what fact would prove this answer?
- Keep review short and specific.

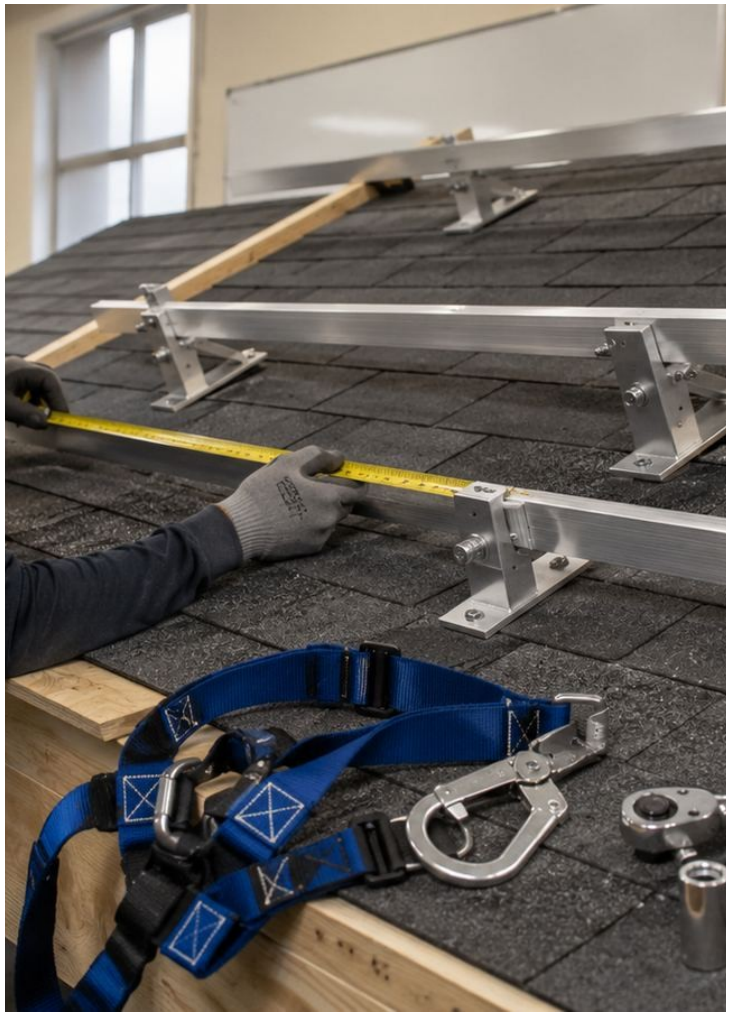


The quiz measures instruction as much as students.

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Core rule

- Plan the install before lifting parts.
- Say it once at the start.
- Bring every debate back to it.
- Use it to reject weak handoffs.



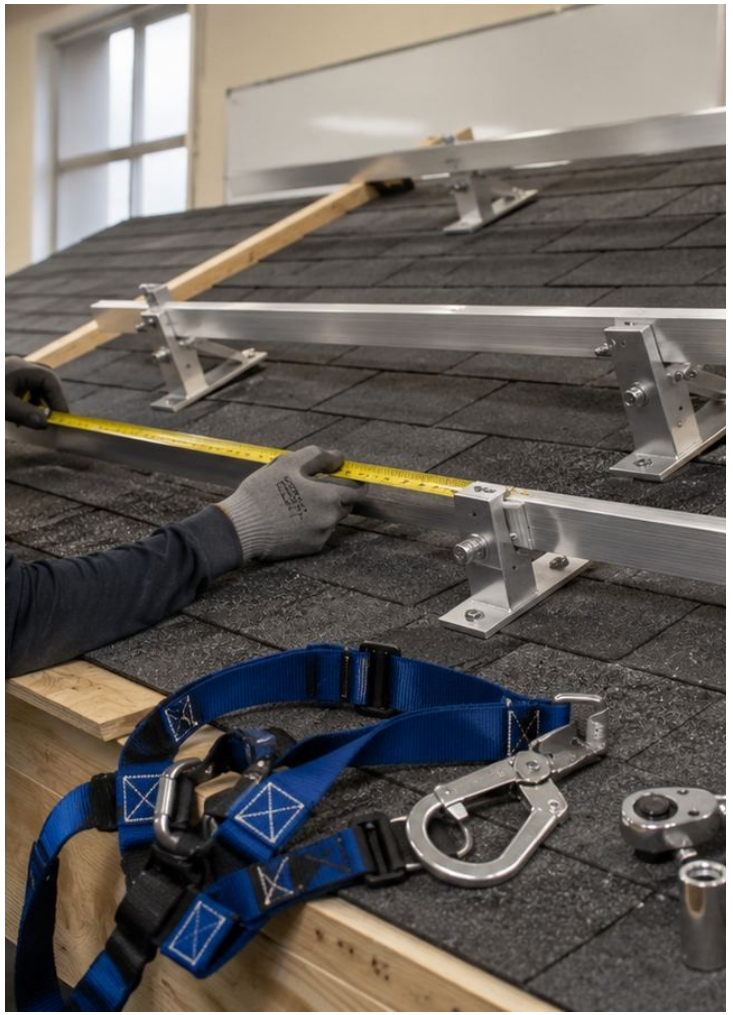
One rule, repeated until reality behaves.

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What students already know

PV101 and RCS are not separate islands.

- PV modules produce DC power.
- Series and parallel change voltage/current behavior.
- Continuity requires de-energized verification.
- Troubleshooting starts from evidence.
- Handoffs protect the next person.

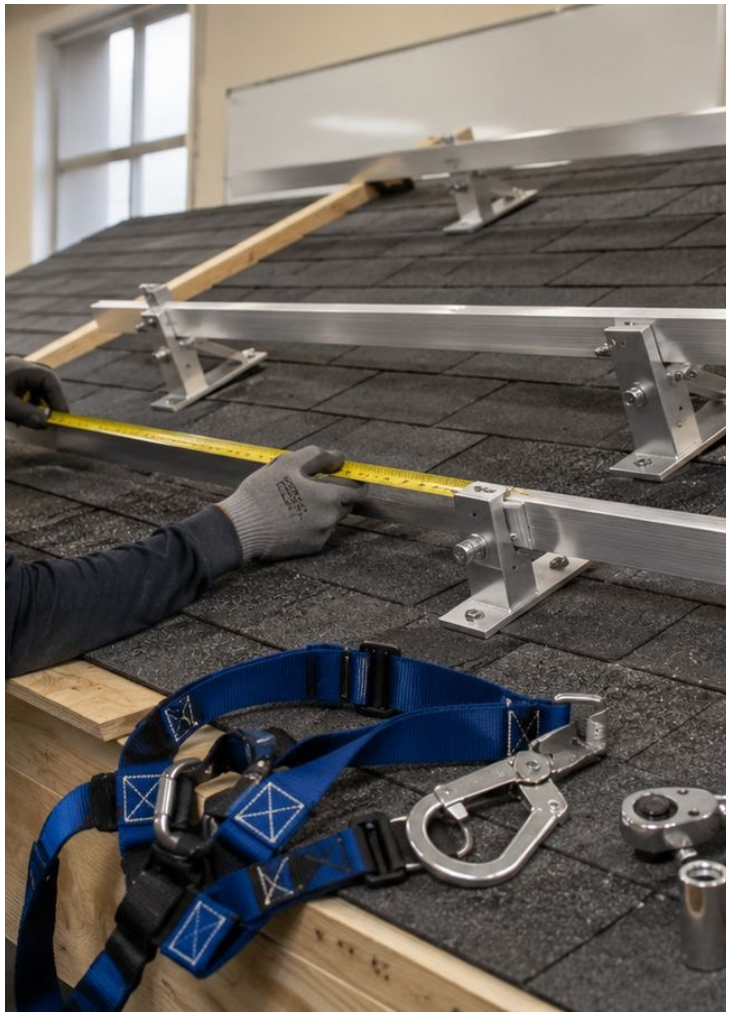


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New judgment today

The skill moves from “can I build it?” to “should this be the design?”

- Define the job.
- Gather the evidence.
- Name the constraint.
- Protect the boundary.
- Record unknowns.



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Energy path and decision boundary



Every arrow is a boundary. Boundaries need ratings, protection, isolation, and documentation.

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Decision checks

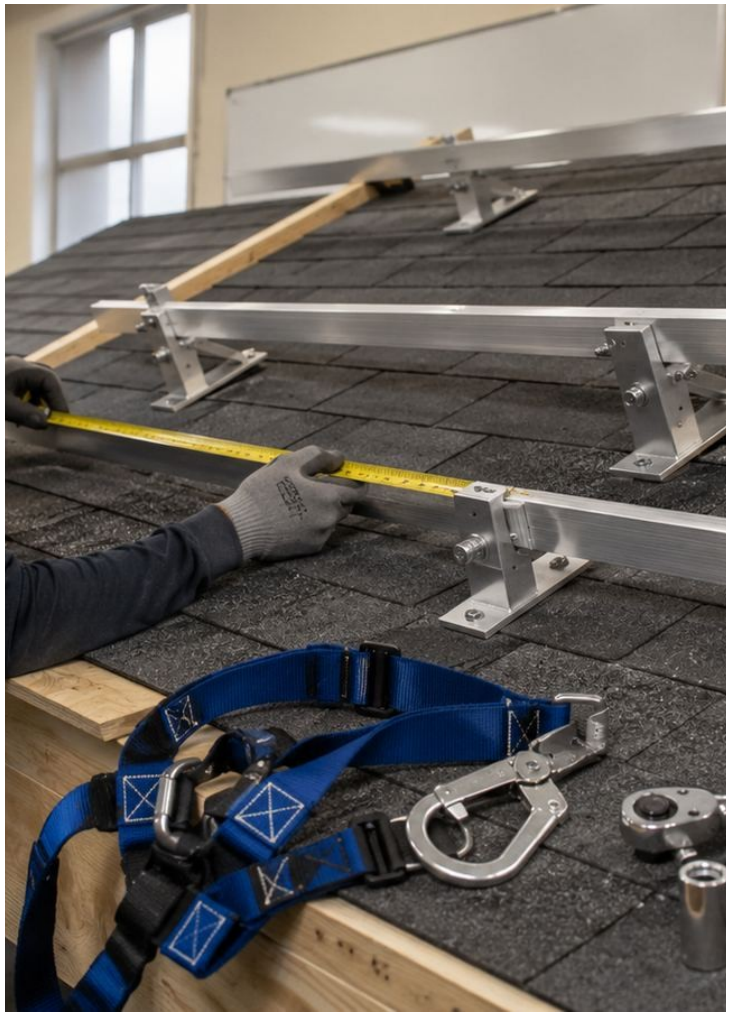
Decision	Evidence	Reject if
Keep a design choice	Measured or observed site/load data	It only sounds good
Reject a shortcut	Safety or equipment boundary	It hides a hazard
Carry unknown forward	Named verification step	It becomes fake certainty
Approve handoff	Next person can act safely	It depends on mind reading

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Guided example

Model one complete thinking path.

- Start with the client/job statement.
- Add one measured or observed fact.
- Make one design decision.
- Name one safety boundary.
- Name one unknown and how to verify it.

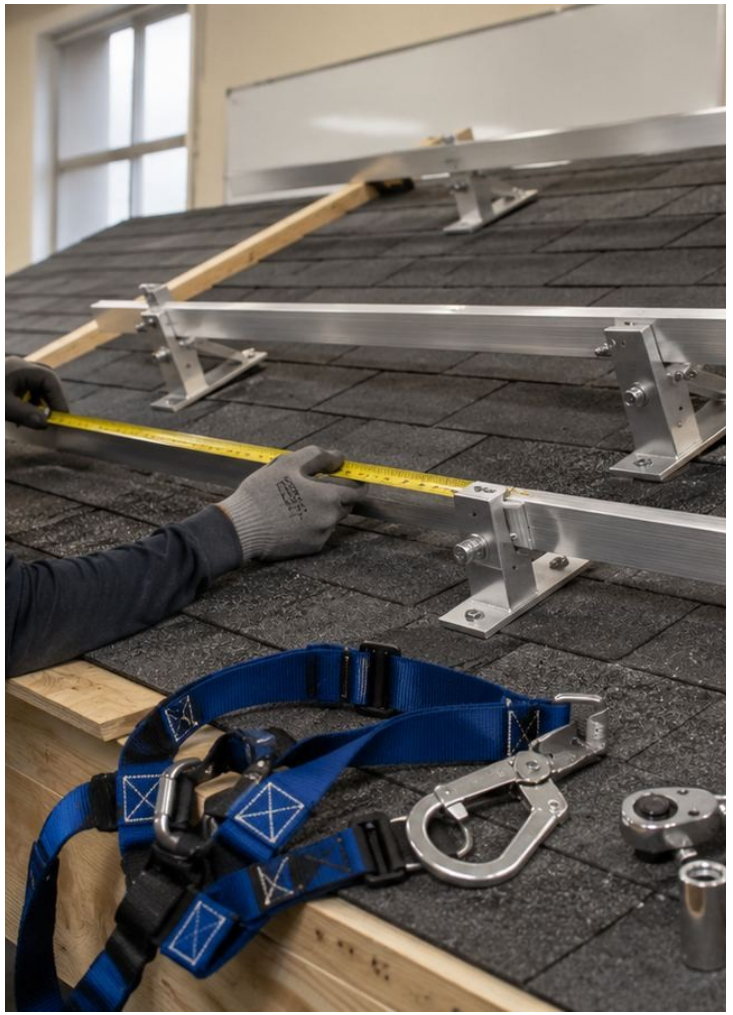


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Team roles

Three teams of two, roles rotate after the first pass.

- Designer: proposes the first path.
- Verifier: challenges evidence and limits.
- Recorder/handoff lead: writes what the next person needs.
- In a pair, one person may hold two roles, then swap.

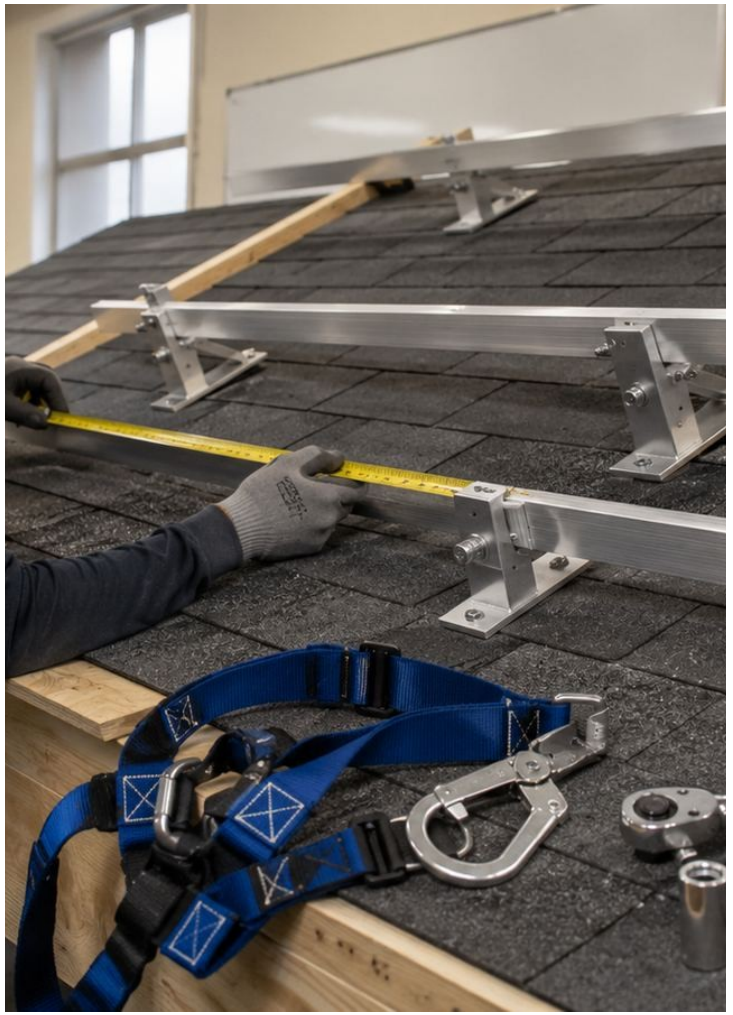


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Lab mission

Install Plan and Job Hazard Analysis

- Teams create a one-day install plan for a training system and defend the safety controls before work begins.



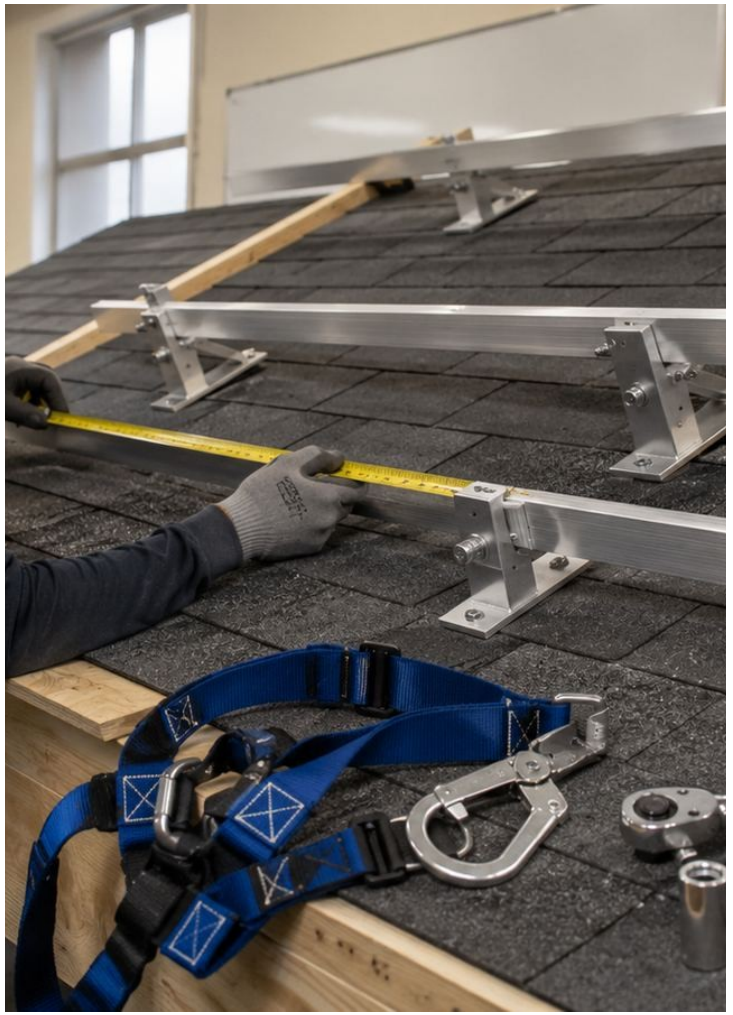
This is the work block. Keep it practical and evidence-based.

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Scenario options

Assign one, or let teams choose if the room can handle democracy today.

- Tabletop PV training kit build: no energized connections until inspection.
- Mock roof rack layout: rails, clamps, wire management, no live module connection.
- Bus conversion equipment board: battery/inverter/controller placement and service clearances.

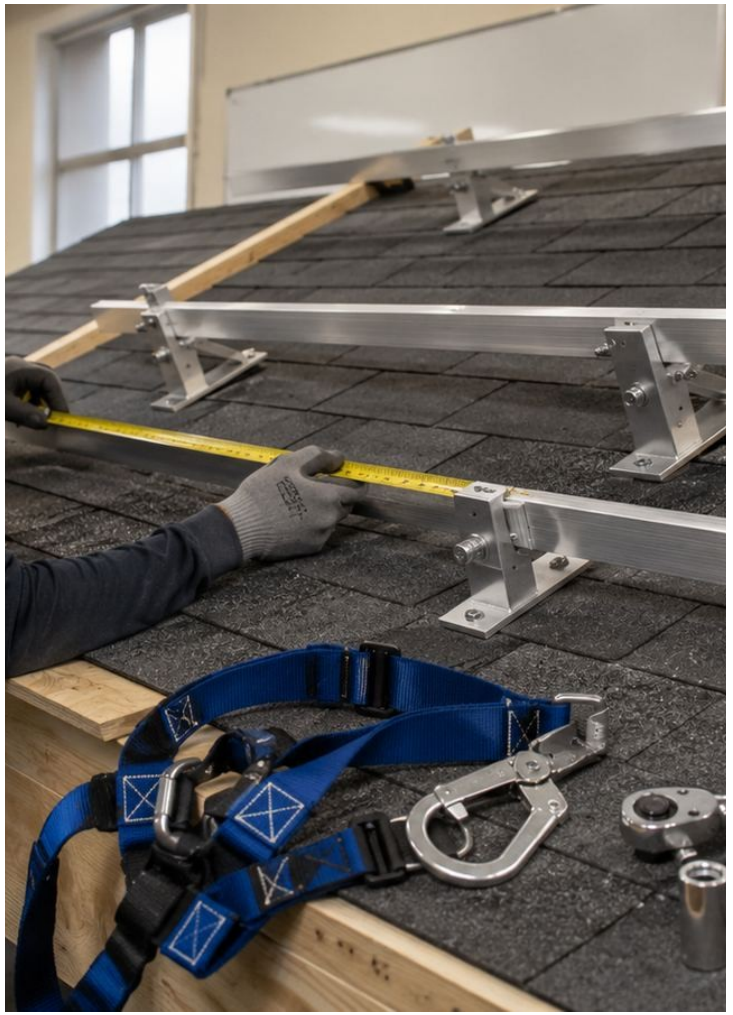


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Lab procedure

Run the same loop every day.

- Choose a scenario and define the scope in one sentence.
- Create a sequenced work plan: setup, layout, mounting/staging, wiring route, checks, cleanup.
- Complete a JHA with hazards, controls, PPE, and stop-work triggers.
- Build a material/tool list and mark missing items.
- Add at least three quality hold points.

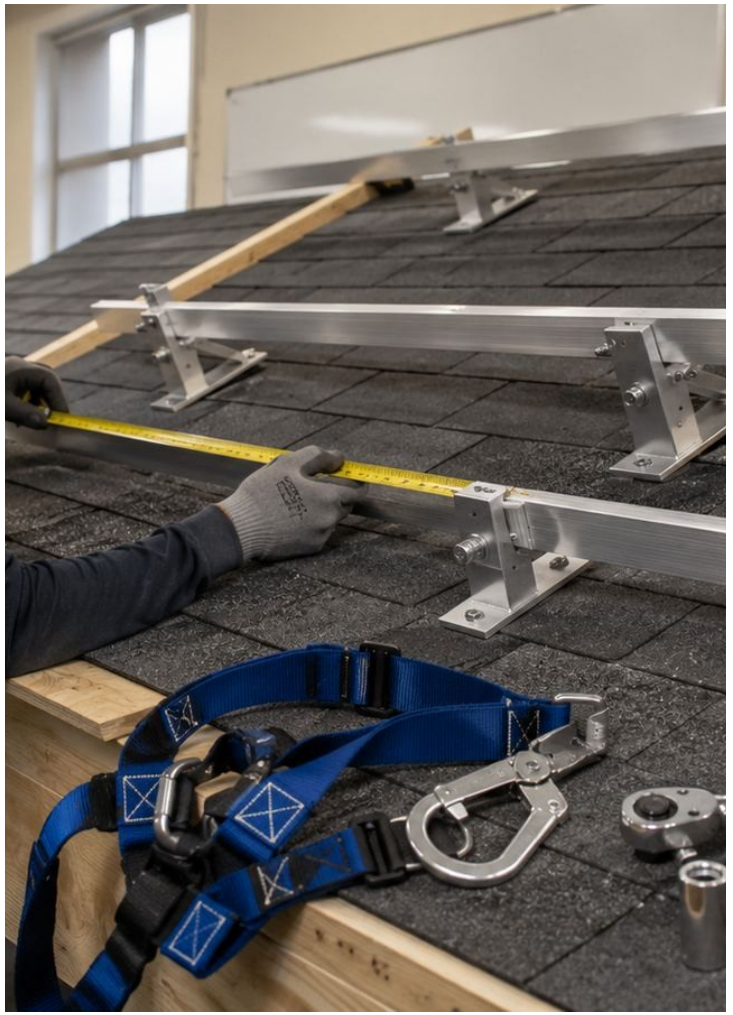


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Acceptance criteria

A complete answer has five parts.

- Match the need.
- Show the path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.



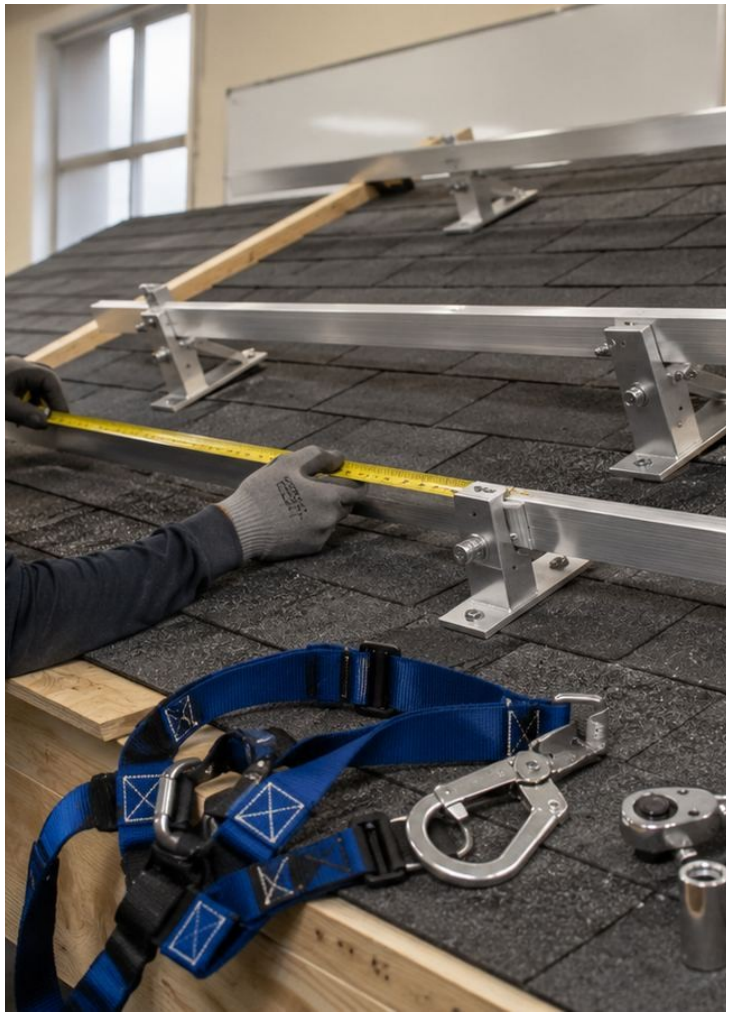
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Safety boundary

No training objective is improved by pretending energy is harmless.

- Use only instructor-approved equipment and procedures.
- Treat PV and batteries as energy sources.
- Stop when measurements do not make sense.
- Follow manufacturer instructions and program safety procedures.

Stop, isolate, verify, ask. The glamorous life.

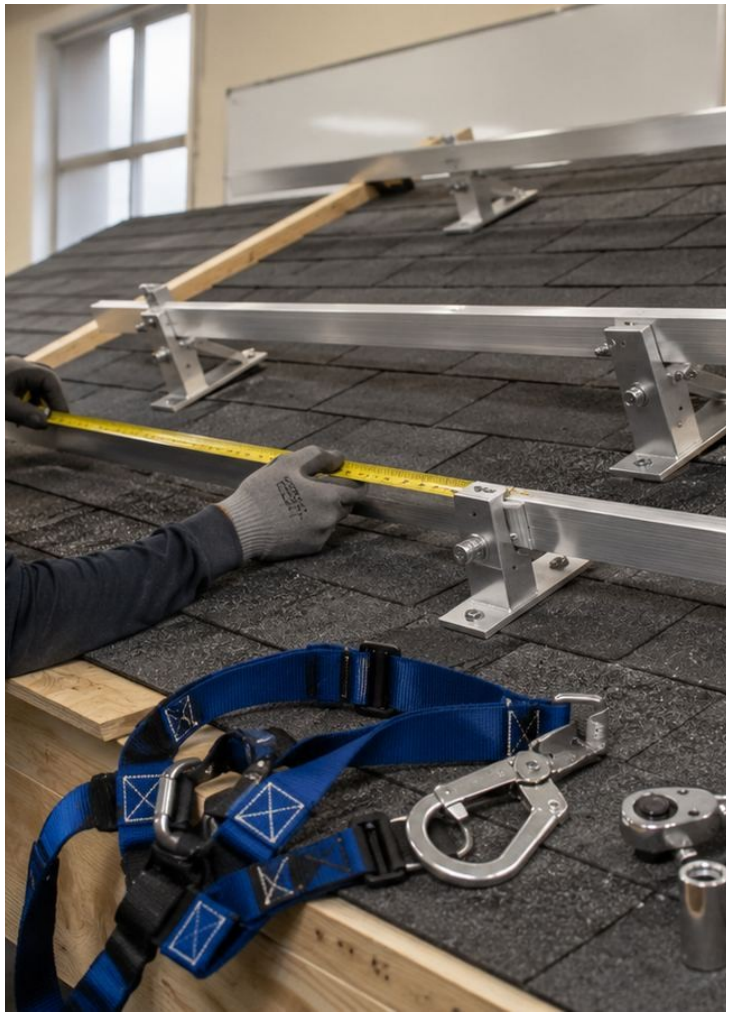


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Instructor probe questions

Use these when teams get vague.

- What evidence supports that?
- What would make this design fail?
- Which boundary protects the person or equipment?
- What unknown are you carrying forward?
- What would the next crew need to know?

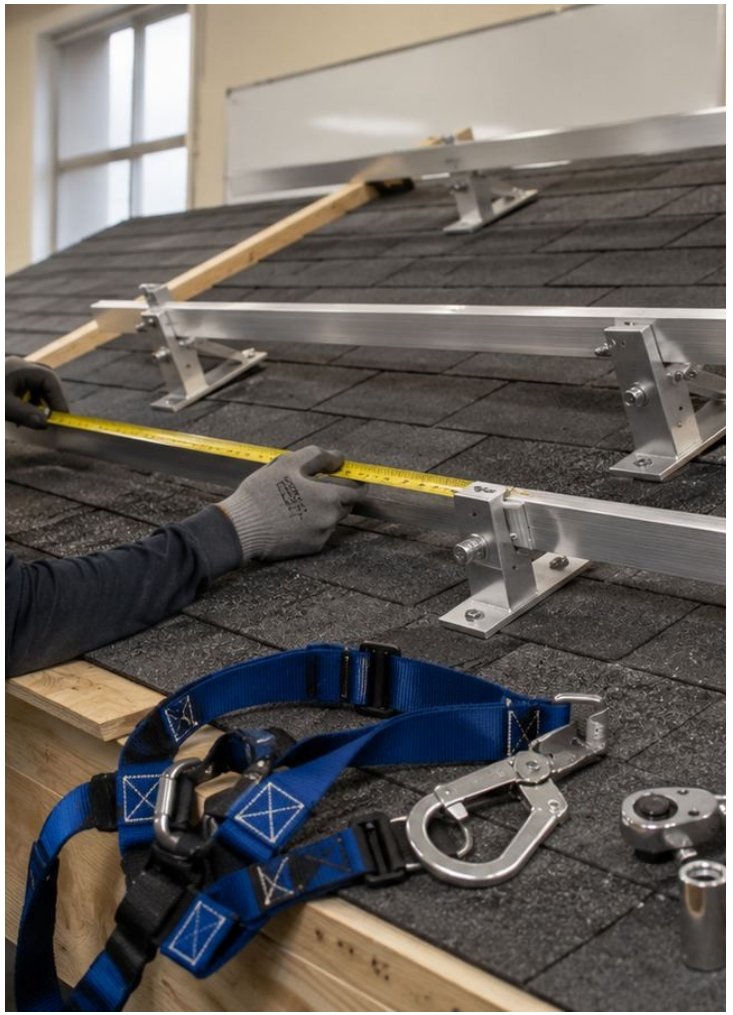


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Peer verification

Peers check the handoff, not the personality of the speaker. Humanity struggles here.

- Can I identify the job?
- Can I follow the path?
- Can I name the constraint?
- Can I see the safety boundary?
- Can I find the unknowns?

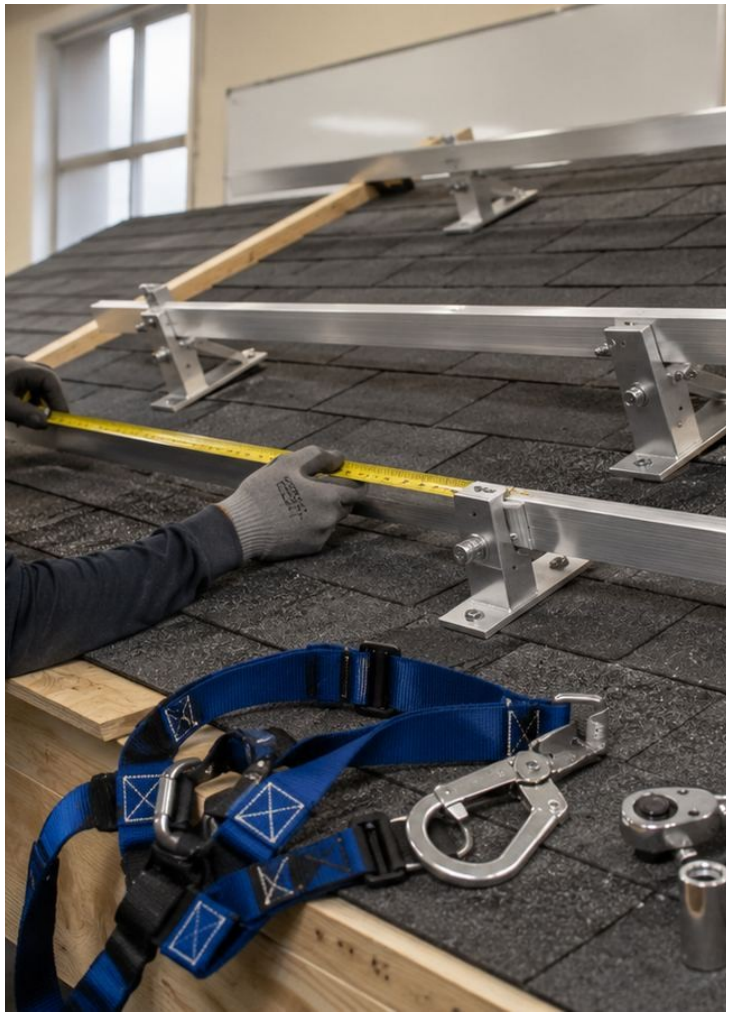


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Common failure modes

Correct these early.

- Component list with no job.
- Guess presented as fact.
- Missing safety boundary.
- No units or runtime.
- Handoff too vague to use.

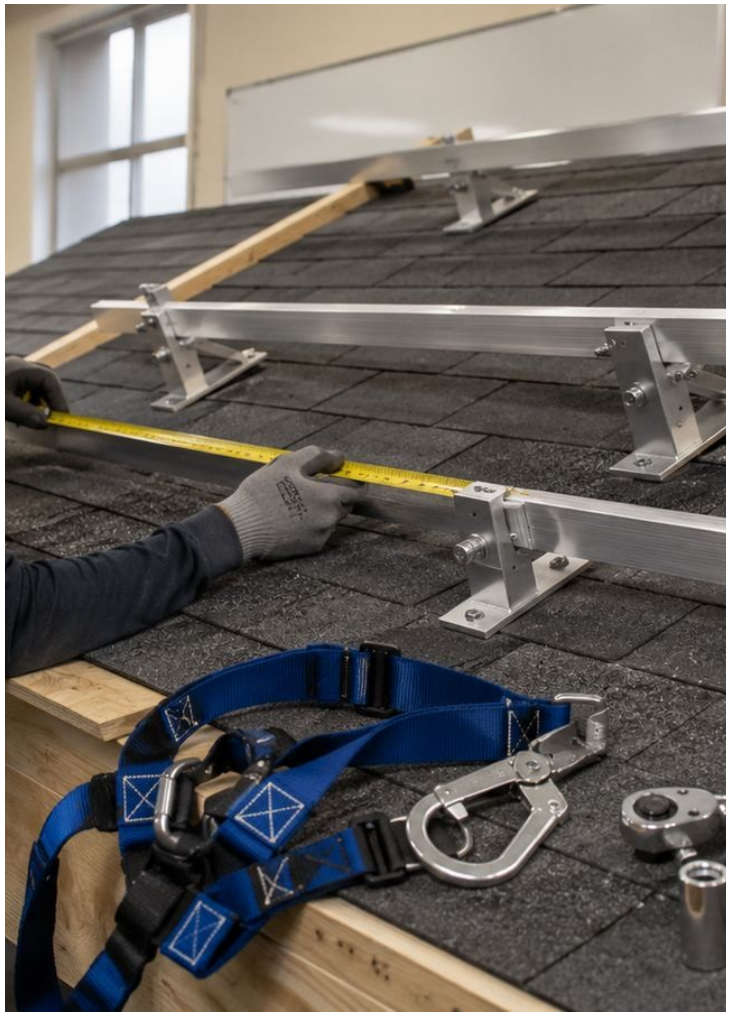


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Evidence language

Require technical sentences.

- Observed: what you saw.
- Measured: value with unit or clear method.
- Assumed: guess being carried forward.
- Verified: checked against source, procedure, or measurement.

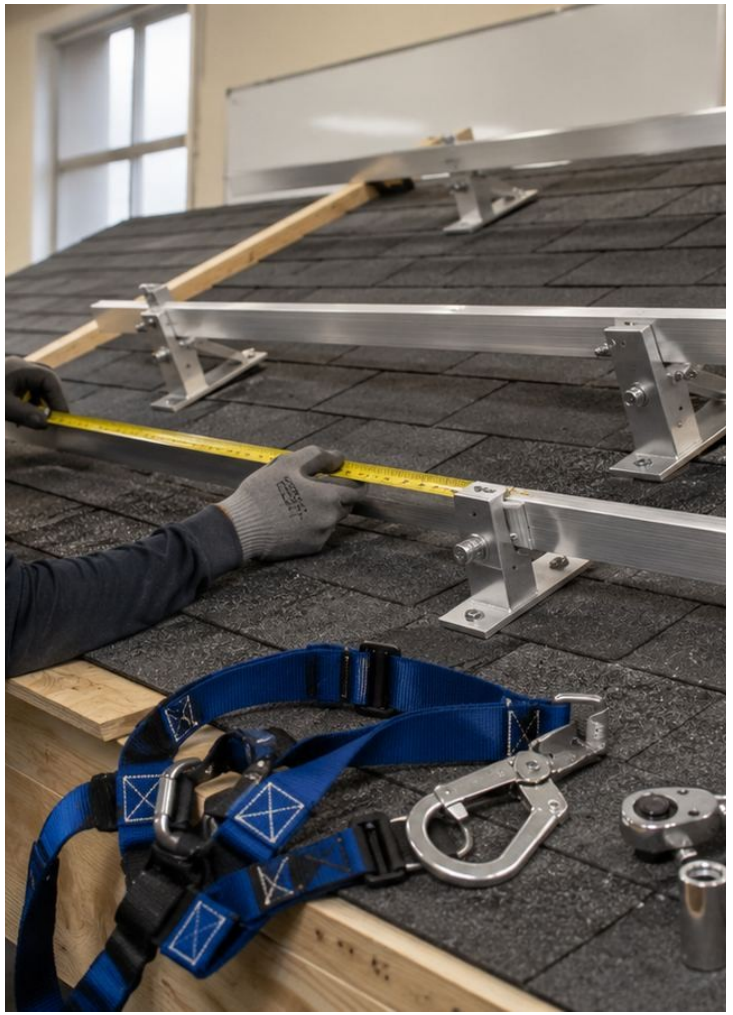


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Handoff structure

Make the next person safer.

- Scope: what job is being solved.
- Decision: what was chosen or rejected.
- Evidence: why.
- Boundary: what must be protected.
- Unknowns: what still needs verification.

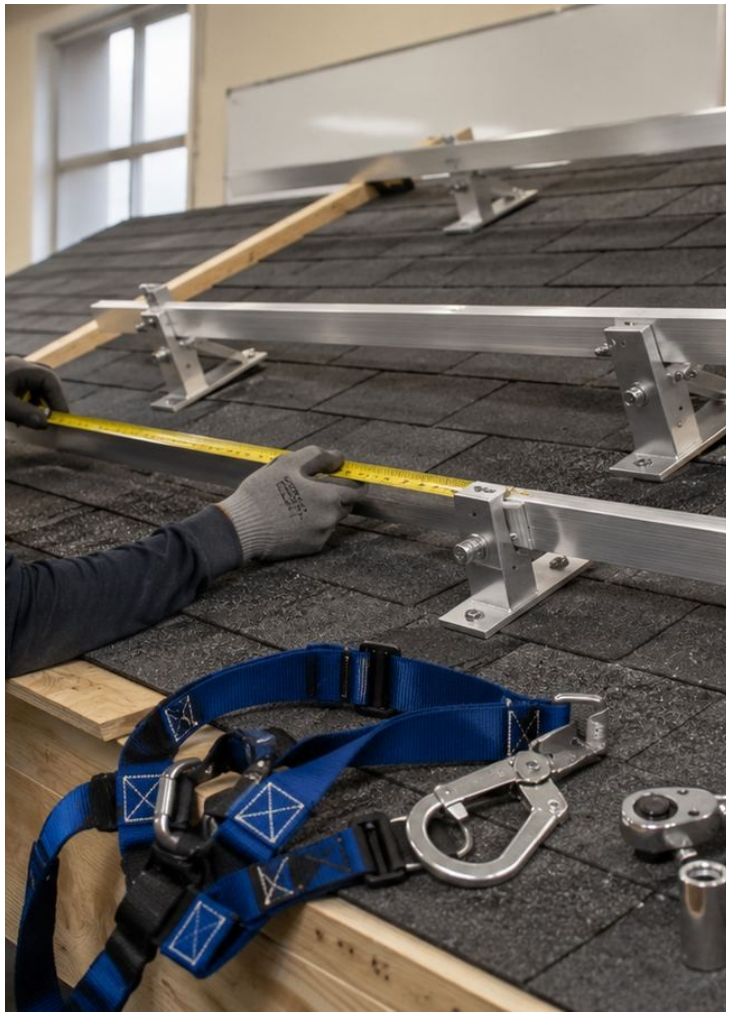


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Exit ticket

Individual close-out

- What is one step your team added that prevents a mistake before it becomes expensive or unsafe?

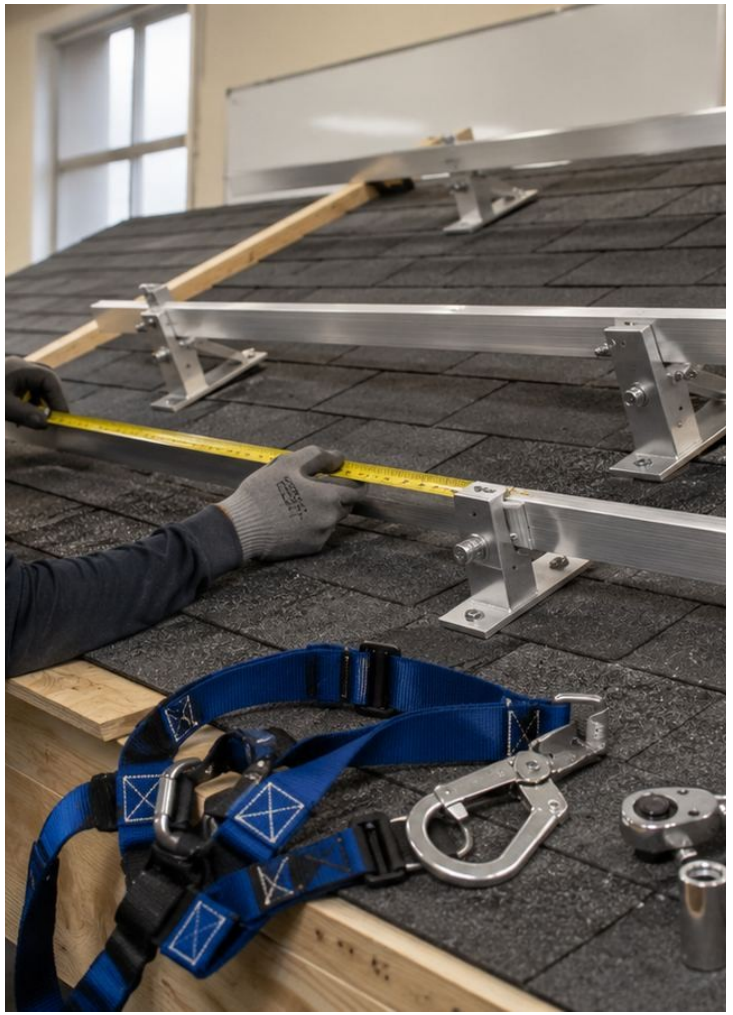


Collect before reset so the day leaves a paper trail. Weirdly useful.

Day-to-day continuity

This is one system, not seven disconnected worksheets.

- PV102 Day 8 uses commissioning and troubleshooting to verify the system and complete a professional handoff.
- Carry forward site notes, load tables, diagrams, and unknowns.
- Do not erase uncertainty. Resolve it.

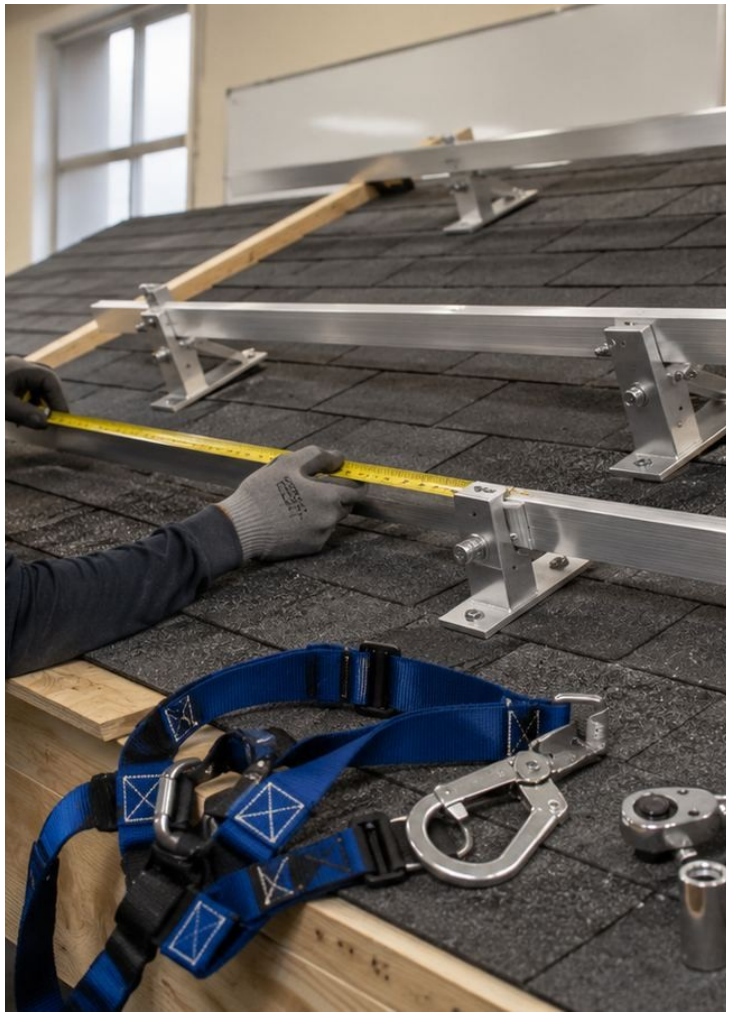


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Final reminder

What professionals do

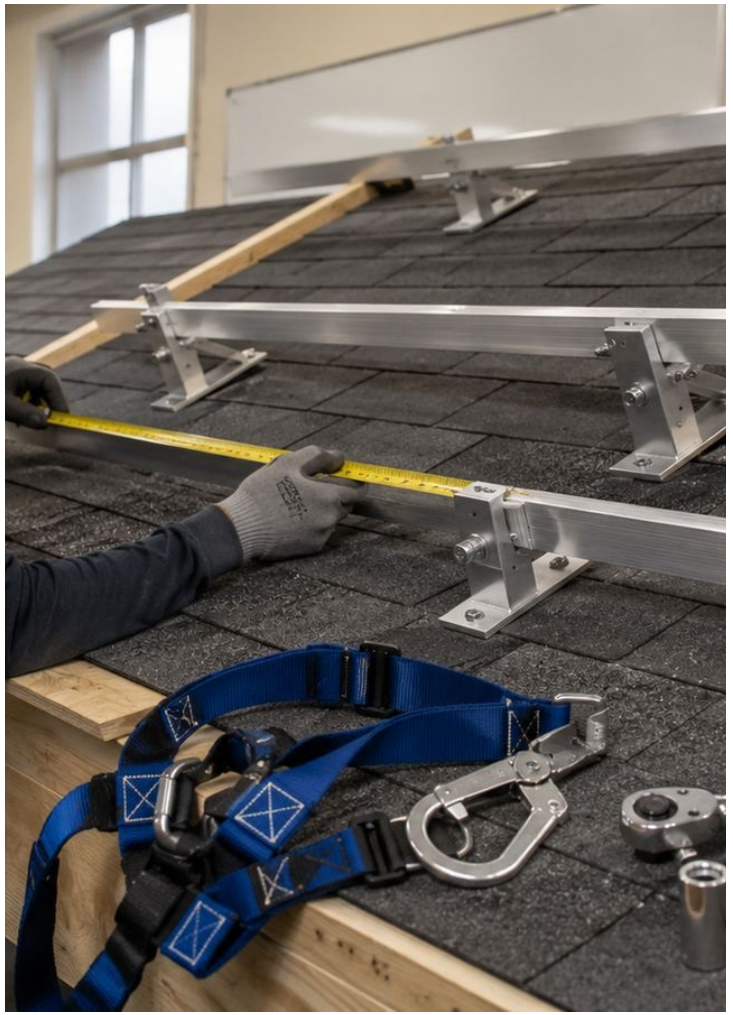
- Define the job.
- Verify the path.
- Protect the boundary.
- Document the truth.



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Close

- Plan the install before lifting parts.



Reset tools. Collect packets. Archive the handoffs.

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